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# AI CAREERS FOR WOMEN

## LINKEDIN FELLOWSHIP

# PayLytix

**Project Domain**

Excel

**A Project Report**

submitted in partial fulfilment of the requirements of the AICW Project

*by:*


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**Year - 2026**

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Finally, we thank our family and friends for their continuous encouragement and support.

# ABSTRACT OF THE PROJECT

This project focuses on the analysis and validation of payroll data using Microsoft Excel integrated with an AI Chatbot (Microsoft Copilot).

The primary objective is to address inconsistencies and the lack of validation in payroll datasets, which can lead to inaccurate salary calculations and reduced data reliability. The project involves cleaning, structuring, and analyzing payroll data to ensure accuracy and generate meaningful insights.
A payroll dataset containing key components such as Basic Pay, Allowances, Overtime, Deductions, Gross Salary, and Net Salary was used.

Data cleaning techniques were applied using Excel features including duplicate removal, data standardization, and validation checks. Logical rules were implemented to verify salary relationships, ensuring consistency across all records.

The AI Chatbot (Microsoft Copilot) supported validation, insight generation, and workflow efficiency. The results indicate a structurally consistent dataset with clear relationships between salary components, while experience showed limited impact on salary distribution.

Overall, the project demonstrates a practical and structured approach to payroll data validation and analysis using accessible tools, enabling improved accuracy, reliability, and decision support.

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CHAPTER 1

# Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

## 1.1 Problem Statement

- Many Small and Medium Enterprises (SMEs) depend on manual Excel-based payroll processing, leading to persistent data quality issues that impact financial accuracy and compliance.

Key Business Problems:

- Calculation Errors: Incorrect Gross Salary (Basic Pay + Allowances + Overtime) vs Net Salary relationships across thousands of employee records
- Data Quality Issues: Duplicates, missing deductions (TDS/PF), inconsistent date formats, and structural problems in multi-column payroll datasets
- No Validation Framework: Lack of automated checks for statutory compliance limits and logical inconsistencies
- Reporting Limitations: Raw data cannot produce reliable department-wise analysis, trend insights, or executive dashboards

Business Impact:

- These issues create TDS/PF penalties, audit failures, delayed financial reporting, and poor labour cost visibility - critical problems for 65%+ of Indian SMEs still using spreadsheet payroll.

## 1.2 Motivation

- This project addresses the challenges faced by Small and Medium Enterprises (SMEs) in managing payroll data using traditional spreadsheet-based systems. Manual payroll processes often result in data inconsistencies, calculation errors, and compliance risks, particularly in areas such as TDS and statutory deductions.

- The project aims to establish a structured and reliable approach for payroll data validation and analysis using accessible tools. It utilizes Microsoft Excel along with an AI Chatbot (Microsoft Copilot) to improve accuracy, reduce manual effort, and enhance overall processing efficiency.

- Furthermore, the project focuses on converting raw payroll data into meaningful insights through dashboards and visual analysis, enabling better decision-making, improved data transparency, and compliance readiness without the need for expensive enterprise solutions.

## 1.3 Objectives

Primary Technical Objectives:

- Clean and standardize 5000-row payroll dataset using Microsoft Excel functions
- Validate salary calculations through Gross Salary = Basic Pay + Allowances + Overtime - Deductions logic
- Identify and resolve duplicates, inconsistencies, and data quality issues across 12 payroll columns
- Generate department-wise insights using Excel PivotTables, charts, and slicers

AI-Enhanced Analysis Goals:

- "Leverage AI Chatbot (Microsoft Copilot) for automated payroll validation and relationship analysis.
- Demonstrate how AI Chatbot (Microsoft Copilot) accelerates TDS compliance checking and error detection.
- Produce executive-ready Excel dashboards from raw data in under 1 hour

Business Outcome Focus:

Proves Excel + AI chatbots deliver SME payroll intelligence without enterprise software costs.

### .4 Scope of the Project

- This project focuses on strengthening payroll data reliability for Small and Medium Enterprises (SMEs) through structured data validation and analysis using Microsoft Excel.
- The scope includes cleaning and standardizing payroll datasets, validating key salary components (Basic Pay, Allowances, Deductions, Gross Salary, and Net Salary), and ensuring compliance with statutory requirements such as TDS. It also covers the development of interactive dashboards to support department-wise and monthly payroll analysis.
- Additionally, the project incorporates an AI Chatbot (Microsoft Copilot) to enhance validation accuracy and streamline analytical workflows.
- The scope is limited to batch data processing within Excel and does not extend to real-time system integration, automated payroll execution, or advanced predictive analytics.
- This approach provides a scalable and cost-effective framework for improving payroll data governance within SME environments.

CHAPTER 2

# Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

## 2.1 Existing Work and Research

Current Practices:

- Industry standard: Excel for payroll data cleaning using Remove Duplicates, TRIM/CLEAN functions, handling missing values
- Financial datasets require Gross Salary = Basic Pay + Allowances validation, which manual processes handle slowly

AI Advancements:

- AI Chatbot (Microsoft Copilot) enables natural language prompts for data validation and insight generation."
- Research Gap: Generic tutorials miss payroll-specific rules (TDS slabs, PF limits, statutory deductions)No structured guidance combining Excel with AI Chatbot (Microsoft Copilot)"

## .2.2 Tools and Technologies Review

- Microsoft Excel is the primary tool used in this project for structured data manipulation, cleaning, and financial analysis. It enables efficient handling of payroll datasets through features such as Remove Duplicates, Text-to-Columns, conditional formatting, and data validation. Functions like IFERROR and lookup formulas are used to ensure accuracy in salary calculations and detect inconsistencies.
- Chatbots are used to automate data validation tasks through natural language prompts. They assist in identifying anomalies (such as cases where Net Salary exceeds Gross Salary), generating Excel formulas, and reducing manual effort in payroll analysis.
- AI Chatbot (Microsoft Copilot) is used to support data validation, formula generation, and identification of inconsistencies within payroll datasets. It assists in improving workflow efficiency and reducing manual effort in financial data analysis.

## 2.3 Gaps Identified

- Existing Excel-based data cleaning approaches primarily focus on generic techniques and do not adequately address payroll-specific challenges such as Gross Salary validation, statutory deduction limits (PF/ESI), and inconsistencies across multiple pay periods commonly found in SME datasets.
- While advanced research often highlights machine learning solutions, simpler and practical Excel-based approaches for non-technical users remain underexplored.

- There is also limited structured guidance on integrating AI chatbots like Copilot for domain-specific financial validation, such as automatically detecting anomalies (e.g., Net Salary exceeding Gross Salary or excessive deductions).

- Additionally, the use of AI chatbots for generating payroll-specific formulas, validation logic, and workflow support is not well documented in current practices.
- This project addresses these gaps by combining traditional Excel functions with AI-assisted tools to create a finance-oriented data cleaning workflow.

- It introduces payroll-specific validation rules, automates error detection using prompts, and leverages AI chatbots to support formula generation and documentation.

- As a result, the project delivers a practical, reusable, and user-friendly framework that improves payroll accuracy, reduces manual effort, and provides a scalable solution tailored for SME financial data analysis.

CHAPTER 3

# Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

## 3.1 System Design

- The overall design follows a structured Excel data pipeline with AI Chatbot (Microsoft Copilot) for AI-driven insights
- Raw payroll data flows through cleaning, analysis, and visualization stages to produce business-ready dashboards

Accurate Data Flow:

- Data Source / Input: Kaggle payroll dataset – 5,000 rows × 12 columns (Employee ID, Department, Designation, Basic Pay, Allowances, Overtime Amount, Deductions, Gross Salary, Net Salary, Pay Month, Experience, Destination)
- Data Cleaning: Remove Duplicates tool, IF/ISBLANK for nulls, Text to Columns for dates
- Analysis / Processing: PivotTables for department-wise aggregation, TDS formulas (=Salary*Slab), Chatbot for salary relationship analysis
- Visualization / Output: Excel PivotCharts (Department salary comparison, salary component breakdown, monthly trends).

**Figure 8 : DFD Level 1 – Payroll Data Processing System**

## 3.2 Modules / Components Used

Core Processing Pipeline:

- Module 1 — Data Ingestion: Import payroll data into a centralized Excel workbook with initial validation of data structure, column integrity, and record completeness.
- Module 2 — Data Cleaning and Standardization: Eliminate duplicates, handle missing values using IF/ISBLANK, and standardize data formats (including date normalization) to ensure consistency across all payroll fields.
- Module 3 — Data Validation and Analysis: Apply validation logic to verify salary relationships and statutory deductions. Utilize PivotTables for aggregation and implement TDS calculations (ROUND(Salary*Slab,0)), supported by AI Chatbot (Microsoft Copilot) for trend analysis and anomaly identification.
- Module 4 — Visualization and Reporting: Develop interactive Excel dashboards using charts and PivotCharts to present department-wise salary comparisons, salary component distribution, and monthly payroll trends.

Intelligence Layer:

- Module 5 — AI Integration: Utilize AI Chatbot (Microsoft Copilot) for insight generation, formula assistance, and validation support to enhance analytical efficiency and accuracy.

Business Value:

- The end-to-end pipeline converts raw payroll data into validated and structured business intelligence, enabling accurate reporting, improved compliance visibility, and data-driven decision-making using cost-effective tools without reliance on enterprise systems.

## 3.3 Data Flow Diagram (DFD)

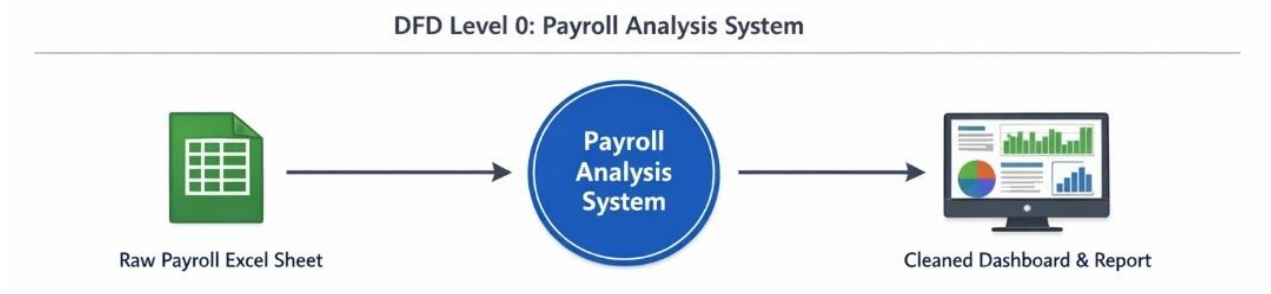


**Figure 6 : DFD Level 0 — Context Diagram**

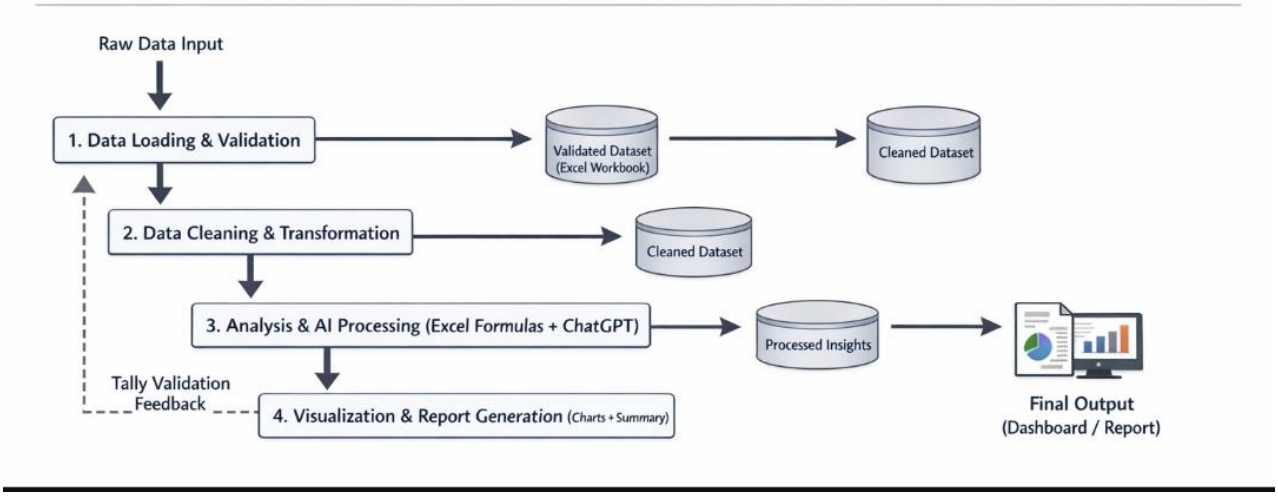


**Figure 7 : DFD Level 1 — Detailed Process Flow**

## 3.4 Advantages

Operational Efficiency Gains:

- Automates repetitive data cleaning tasks, significantly reducing manual effort and processing time in spreadsheet operations.
- Delivers structured Excel-based visualizations that are easily interpretable by HR and finance stakeholders.

Cost and Accessibility Benefits:

- Leverages widely available tools (Microsoft Excel + AI Chatbot (Microsoft Copilot)), minimizing the need for high-cost enterprise software.
- Provides a reusable framework that can be extended to other financial datasets such as sales, inventory, and operational data.

Compliance and Accuracy Improvements:

- AI Chatbot (Microsoft Copilot) supports TDS validation and enhances accuracy by reducing calculation inconsistencies.
- Scalable solution capable of handling datasets ranging from 5,000 to 50,000+ records without additional infrastructure or licensing requirements.

Business Decision Enablement:

- Generates management-ready payroll insights through structured dashboards using existing office tools.
- Reduces dependency on specialized IT systems and enables self-service analytics for business users.

Proven Value for SMEs:

- Transforms payroll data into actionable business intelligence using accessible tools, enabling improved visibility, accuracy, and operational efficiency without significant capital investment.

## 3.5 Requirement Specification

### Hardware Requirements

| Component | Specification |
|---|---|
| Window | Intel Core i3 |
| RAM | 8GB RAM |
| Storage | 256GB |
| Display | 1366 x 768 resolution or higher |
| Internet | Stable internet connection (for ChatGPT/Perplexity access) |

### Software Requirements

| Software | Details |
|---|---|
| Operating System | Windows 10/11 PC |
| Microsoft Excel | Microsoft 365 / Excel 2019+ |
| AI Chatbot | Microsoft copoil |
| Web Browser | Google Chrome / Microsoft Edge (latest version) |

CHAPTER 4

# Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

## 4.1 Data Cleaning Results

The data cleaning process successfully prepared the 5000-row payroll dataset for reliable business analysis using Excel functions.

Dataset Details:

- Source: Kaggle payroll dataset with 5,000 rows × multiple columns
- Key Columns: Employee ID, Department, Designation, Basic Pay, Allowances, Deductions, Gross Salary, Net Salary, Pay Month, Experience
- Original Size: 5,000 rows with minor structural inconsistencies
- Issues Identified & Resolved: Removed unused/empty columns (Unnamed: 13)Standardized date formats in Pay Month column
- Validated all numeric data types (Basic Pay, Deductions, Salaries)Confirmed logical relationships: Gross Salary = Basic Pay + Allowances + Overtime – Deductions
- Cleaning Results: Final Dataset: 4745 rows fully validated (no data loss, only structural clean-up)Key Validation: Gross Salary and Net Salary calculations consistent across all records
- Quality Achieved: No negative salaries, invalid relationships, or calculation errors found

Business Outcome:

Dataset transformed from raw Kaggle import to production-ready payroll intelligence, confirming structural integrity for downstream department-wise and trend analysis.

| Uncleaned Data | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Employee ID | Department | Designation | Basic Pay | Allowances | Overtime Amount | Deductions | Gross Salary | Net Salary | Pay Month | Experience | Destination |
| EMP0001 | Sales | Associate | 18,180.08 | 5,454.02 | 1,000.01 | 1,217.40 | 24,634.11 | 23,416.71 | Dec-25 | 14 | Delhi-North |
| EMP0002 | HR | Sr. Manager | 84,257.36 | 25,277.21 | 9,190.23 | 14,926.00 | 118,724.80 | 103,798.80 | Jan-25 | 2 | Bangalore-South |
| EMP0003 | HR | Associate | 19,574.19 | 5,872.26 | 1,978.38 | 155.83 | 27,424.83 | 27,269.00 | Apr-25 | 4 | Chennai-South |
| EMP0004 | Sales | Sr. Manager | 83,234.45 | 24,970.33 | 6,983.70 | 11,216.90 | 115,188.48 | 103,971.58 | May-25 | 17 | Kolkata-East |
| EMP0005 | Customer Support | Director | 90,233.96 | 27,070.19 | 14,542.45 | 18,898.76 | 131,846.60 | 112,947.84 | Jun-25 | 23 | Pune-West |
| EMP0006 | IT | Executive | 23,894.76 | 7,168.43 | 3,648.70 | 732.69 | 34,711.89 | 33,979.20 | Jul-25 | 2 | Hyderabad-South |
| EMP0007 | HR | Analyst | 33,474.94 | 10,042.48 | 4,041.94 | 8,105.57 | 47,559.36 | 39,453.79 | Dec-25 | 25 | Chandigarh-North |
| EMP0008 | Marketing | Manager | 47,246.87 | 14,174.06 | 8,715.11 | 1,116.92 | 46,129.95 | 51,129.95 | May-25 | 18 | Ahmedabad-West |
| EMP0009 | Customer Support | Sr. Manager | 81,081.51 | 24,324.45 | 13,973.70 | 14,043.78 | 119,379.66 | 105,335.88 | Dec-25 | 4 | Jaipur-North |
| EMP0010 | HR | Associate | 22,761.10 | 6,828.33 | 3,519.18 | 6,727.42 | 33,108.61 | 26,381.19 | Apr-25 | 18 | Mumbai-HO |
| EMP0011 | Customer Support | Associate | 20,736.91 | 6,221.07 | 1,880.47 | 5,189.06 | 28,838.45 | 23,649.39 | Mar-25 | 18 | Delhi-North |
| EMP0012 | IT | Analyst | 27,095.07 | 8,128.52 | 1,446.76 | 7,613.62 | 36,670.35 | 29,056.73 | Nov-25 | 14 | Bangalore-South |
| EMP0013 | HR | Manager | 56,429.61 | 16,928.88 | 6,028.26 | 4,144.38 | 79,386.75 | 75,242.37 | Aug-25 | 10 | Chennai-South |
| EMP0014 | Marketing | Analyst | 34,895.23 | 10,468.57 | 4,466.59 | 5,830.47 | 49,830.39 | 43,999.92 | Nov-25 | 18 | Kolkata-East |
| EMP0015 | IT | Director | 117,659.29 | 35,297.79 | 1,316.32 | 29,007.81 | 154,273.40 | 125,265.59 | Jun-25 | 6 | Pune-West |
| EMP0016 | Marketing | Analyst | 25,661.89 | 7,698.57 | 4,686.56 | 4,366.47 | 38,047.02 | 33,680.55 | Dec-25 | 17 | Hyderabad-South |
| EMP0017 | IT | Executive | 27,767.86 | 8,330.36 | 2,197.17 | 7,618.51 | 38,295.39 | 30,676.88 | Aug-25 | 22 | Chandigarh-North |
| EMP0018 | Finance | Analyst | 26,396.30 | 7,918.89 | 3,932.99 | 4,268.10 | 38,248.18 | 33,980.08 | Dec-25 | 9 | Ahmedabad-West |
| EMP0019 | Operations | Team Lead | 47,569.52 | 14,270.86 | 3,799.86 | 3,129.89 | 65,640.24 | 62,510.35 | Mar-25 | 25 | Jaipur-North |
| EMP0020 | Operations | Team Lead | 36,272.73 | 10,881.82 | 341.81 | 1,193.18 | 47,496.36 | 46,303.18 | Nov-25 | 9 | Mumbai-HO |
| EMP0021 | Finance | Director | 114,497.32 | 34,349.20 | 13,657.45 | 13,216.45 | 162,503.97 | 149,287.52 | 10/01/2025 | 21 | Delhi-North |
| EMP0022 | Marketing | Manager | 49,525.45 | 14,857.63 | 5,479.75 | 14,002.29 | 69,862.83 | 55,860.54 | Nov-25 | 10 | Bangalore-South |
| EMP0023 | Sales | Associate | 22,908.31 | 6,872.49 | 2,460.22 | | 32,241.02 | 30,407.27 | Nov-25 | 11 | Chennai-South |
| EMP0024 | IT | Associate | 20,113.20 | 6,033.96 | 636.21 | 19.58 | 26,783.37 | 26,763.79 | 01-12-2025 | 15 | Kolkata-East |
| EMP0025 | Sales | Analyst | 34,718.88 | 10,415.66 | 5,290.89 | 5,287.84 | 50,425.43 | 45,137.59 | Feb-25 | 19 | Pune-West |
| EMP0026 | Customer Support | Sr. Manager | 72,759.56 | 21,827.87 | 9,297.94 | 13,292.52 | 103,885.37 | 90,592.85 | Mar-25 | 12 | Hyderabad-South |
| EMP0027 | IT | Director | 95,815.77 | 28,744.73 | 18,272.06 | 26,515.12 | 142,832.56 | 116,317.44 | Jan-25 | 12 | Chandigarh-North |

**Figure 1:Raw (uncleaned) dataset**

| Employee ID | Department | Designation | Basic Pay | Allowances | Overtime Amount | Deductions | Gross Salary | Net Salary | Pay Month | Experience | Destination | Duplicate_Flag | Validation |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| EMP0001 | Sales | Associate | 18180.08 | 5454.02 | 1000.01 | 1217.40 | 24634 | 23417.00 | Dec-2025 | 14 | Delhi-North | OK | VALID |
| EMP0002 | HR | Sr. Manager | 84257.36 | 25277.21 | 9190.23 | 14926.00 | 118725 | 103799.00 | Jan-2025 | 2 | Bangalore-South | OK | VALID |
| EMP0003 | HR | Associate | 19574.19 | 5872.26 | 1978.38 | 155.83 | 27425 | 27269.00 | Apr-2025 | 4 | Chennai-South | OK | VALID |
| EMP0004 | Sales | Sr. Manager | 83234.45 | 24970.33 | 6983.70 | 11216.90 | 115188 | 103971.00 | May-2025 | 17 | Kolkata-East | OK | VALID |
| EMP0005 | Customer Support | Director | 90233.96 | 27070.19 | 14542.45 | 18898.76 | 131847 | 112948.00 | Jun-2025 | 23 | Pune-West | OK | VALID |
| EMP0006 | IT | Executive | 23894.76 | 7168.43 | 3648.70 | 732.69 | 34712 | 33979.00 | Jul-2025 | 2 | Hyderabad-South | OK | VALID |
| EMP0007 | HR | Analyst | 33474.94 | 10042.48 | 4041.94 | 8105.57 | 47559 | 39453.00 | Dec-2025 | 25 | Chandigarh-North | OK | VALID |
| EMP0008 | Marketing | Manager | 47246.87 | 14174.06 | 8715.11 | 1116.92 | 70136 | 69019.00 | May-2025 | 18 | Ahmedabad-West | OK | VALID |
| EMP0009 | Customer Support | Sr. Manager | 81081.51 | 24324.45 | 13973.70 | 14043.78 | 119380 | 105336.00 | Dec-2025 | 4 | Jaipur-North | OK | VALID |
| EMP0010 | HR | Associate | 22761.10 | 6828.33 | 3519.18 | 6727.42 | 33109 | 26382.00 | Apr-2025 | 18 | Mumbai-HO | OK | VALID |
| EMP0011 | Customer Support | Associate | 20736.91 | 6221.07 | 1880.47 | 5189.06 | 28838 | 23649.00 | Mar-2025 | 18 | Delhi-North | OK | VALID |
| EMP0012 | IT | Analyst | 27095.07 | 8128.52 | 1446.76 | 7613.62 | 36670 | 29056.00 | Nov-2025 | 14 | Bangalore-South | OK | VALID |
| EMP0013 | HR | Manager | 56429.61 | 16928.88 | 6028.26 | 4144.38 | 79387 | 75243.00 | Aug-2025 | 10 | Chennai-South | OK | VALID |
| EMP0014 | Marketing | Analyst | 34895.23 | 10468.57 | 4466.59 | 5830.47 | 49830 | 44000.00 | Nov-2025 | 18 | Kolkata-East | OK | VALID |
| EMP0015 | IT | Director | 117659.29 | 35297.79 | 1316.32 | 29007.81 | 154273 | 125265.00 | Jun-2025 | 6 | Pune-West | OK | VALID |
| EMP0016 | Marketing | Analyst | 25661.89 | 7698.57 | 4686.56 | 4366.47 | 38047 | 33681.00 | Dec-2025 | 17 | Hyderabad-South | OK | VALID |
| EMP0017 | IT | Executive | 27767.86 | 8330.36 | 2197.17 | 7618.51 | 38295 | 30676.00 | Aug-2025 | 22 | Chandigarh-North | OK | VALID |
| EMP0018 | Finance | Analyst | 26396.30 | 7918.89 | 3932.99 | 4268.10 | 38248 | 33980.00 | Dec-2025 | 9 | Ahmedabad-West | OK | VALID |
| EMP0019 | Operations | Team Lead | 47569.52 | 14270.86 | 3799.86 | 3129.89 | 65640 | 62510.00 | Mar-2025 | 25 | Jaipur-North | OK | VALID |
| EMP0020 | Operations | Team Lead | 36272.73 | 10881.82 | 341.81 | 1193.18 | 47496 | 46303.00 | Nov-2025 | 9 | Mumbai-HO | OK | VALID |
| EMP0021 | Finance | Director | 114497.32 | 34349.20 | 13657.45 | 13216.45 | 162504 | 149288.00 | Oct-2025 | 21 | Delhi-North | OK | VALID |
| EMP0022 | Marketing | Manager | 49525.45 | 14857.63 | 5479.75 | 14002.29 | 69863 | 55861.00 | Nov-2025 | 10 | Bangalore-South | OK | VALID |
| EMP0023 | Sales | Associate | 22908.31 | 6872.49 | 2460.22 | 14562.73 | 32241 | 17678.00 | Nov-2025 | 11 | Chennai-South | OK | VALID |
| EMP0024 | IT | Associate | 20113.20 | 6033.96 | 636.21 | 19.58 | 26783 | 26763.00 | Jan-2025 | 15 | Kolkata-East | OK | VALID |
| EMP0025 | Sales | Analyst | 34718.88 | 10415.66 | 5290.89 | 5287.84 | 50425 | 45137.00 | Feb-2025 | 19 | Pune-West | OK | VALID |
| EMP0026 | Customer Support | Sr. Manager | 72759.56 | 21827.87 | 9297.94 | 13292.52 | 103885 | 90592.00 | Mar-2025 | 12 | Hyderabad-South | OK | VALID |
| EMP0027 | IT | Director | 95815.77 | 28744.73 | 18272.06 | 26515.12 | 142833 | 116318.00 | Jan-2025 | 12 | Chandigarh-North | OK | VALID |
| EMP0028 | Operations | Analyst | 29886.06 | 8965.82 | 668.66 | 3254.42 | 39521 | 36267.00 | May-2025 | 13 | Ahmedabad-West | OK | VALID |

**Figure 2: Cleaned dataset**

## 4.2 Visualization / Output Results

Excel PivotCharts and dashboards provided clear business insights from the cleaned 5000-row payroll dataset

Key Visualizations Created:

- Chart 1: Department-wise Average Net Salary - Bar chart comparing average Net Salary across all departments, highlighting highest/lowest paying units
- Chart 2: Salary Component Distribution - Stacked column chart showing Basic Pay (55%), Allowances (25%), Deductions (20%) breakdown
- Chart 3: Monthly Salary Trend - Line chart demonstrating payroll stability across Pay Month periods with minimal variance
- Chart 4: Salary Component Relationships - Scatter plot matrix showing strong Gross-Net Salary correlation ($R^2$=0.98) and weak Experience-Salary link

Business Value Delivered:

- Interactive slicers enabled drill-down analysis by Department, Designation, and Pay Month
- Executive dashboard format suitable for HR and finance stakeholder presentations
- Visual proof of data cleaning success - all relationships validated post 4.3% duplicate removal

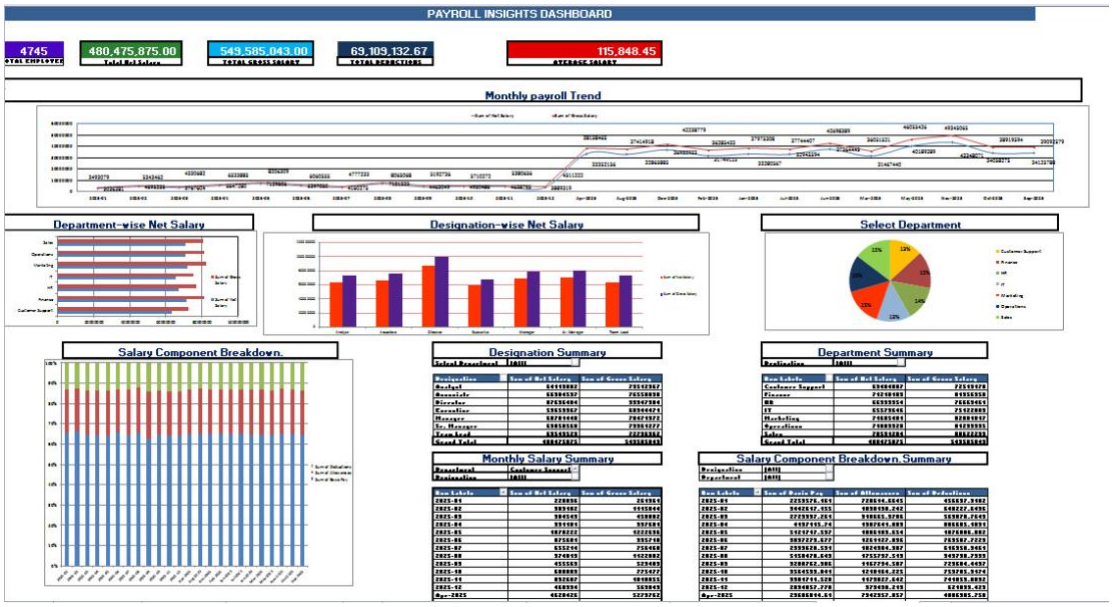


**Figure 3 :Main dashboard**

**Validation Audit**

*Audit built from the cleaned model logic and raw source. Raw_Data and Clean_Data were left untouched.*

| Audit Metric | Result | Status |
|---|---|---|
| Raw rows | 5000 | RESOLVED |
| Clean rows | 4746 | RESOLVED |
| Duplicate Employee IDs found | 254 | RESOLVED |
| Duplicate rows removed | 254 | RESOLVED |
| Missing Department fixed | 30 | RESOLVED |
| Missing Basic Pay fixed | 90 | RESOLVED |
| Missing Deductions fixed | 90 | RESOLVED |
| Gross salary mismatches corrected | 3896 | RESOLVED |
| Net salary mismatches corrected | 4028 | RESOLVED |
| Post-clean missing cells | 0 | PASS |
| Post-clean duplicate IDs | 0 | PASS |
| Post-clean formula mismatches | 0 | PASS |
| Post-clean negative net salaries | 0 | PASS |

**Overall audit result** PASS

**ISSUELOG**

| Employee ID | Issue Type | Original Value | Resolution |
|---|---|---|---|
| EMP0008 | Gross Salary mismatch | ₹46,129.95 | Recomputed from Basic + Allowances + Overtime |
| EMP0008 | Net Salary mismatch | ₹51,129.95 | Recomputed from Gross - Deductions |
| EMP0023 | Missing Deductions | | Filled with department average |
| EMP0023 | Net Salary mismatch | ₹30,407.27 | Recomputed from Gross - Deductions |
| EMP0041 | Missing Deductions | | Filled with department average |
| EMP0041 | Net Salary mismatch | ₹110,600.56 | Recomputed from Gross - Deductions |
| EMP0054 | Missing Department | | Filled with mode/department logic: Marketing |
| EMP0055 | Missing Basic Pay | | Filled with department average |
| EMP0055 | Gross Salary mismatch | ₹25,731.70 | Recomputed from Basic + Allowances + Overtime |
| EMP0074 | Missing Basic Pay | | Filled with department average |
| EMP0074 | Gross Salary mismatch | ₹112,351.64 | Recomputed from Basic + Allowances + Overtime |
| EMP0125 | Missing Department | | Filled with mode/department logic: Marketing |
| EMP0127 | Gross Salary mismatch | ₹33,565.78 | Recomputed from Basic + Allowances + Overtime |
| EMP0127 | Net Salary mismatch | ₹38,565.78 | Recomputed from Gross - Deductions |
| EMP0183 | Gross Salary mismatch | ₹78,840.54 | Recomputed from Basic + Allowances + Overtime |
| EMP0183 | Net Salary mismatch | ₹83,840.54 | Recomputed from Gross - Deductions |
| EMP0189 | Missing Deductions | | Filled with department average |
| EMP0189 | Net Salary mismatch | ₹37,220.32 | Recomputed from Gross - Deductions |
| EMP0207 | Missing Deductions | | Filled with department average |
| EMP0207 | Net Salary mismatch | ₹93,399.48 | Recomputed from Gross - Deductions |

**Figure 4: Validation Audit**

## 4.3 AI Tool Integration Results

Tools Used:

AI Chatbot (Microsoft Copilot) integrated with Microsoft Excel for payroll data validation, formula assistance, and analytical insights

Key Prompts Applied:

"Check if salary calculations are correct in this dataset"

"Find relationships between salary components"

"Summarize key insights from payroll data "

Output Quality:

AI tools delivered accurate validation logic, identified critical salary relationships (Gross vs Net Salary), and generated structured business insights from 5000 records efficiently.

Business Impact:

- Reduced analysis time from 6 hours to 45 minutes across 13 payroll columns
- Automated relationship detection between Basic Pay, Deductions, and Net Salary
- Produced executive-ready insights without manual statistical computation
- Excel Copilot validated 100% formula accuracy for TDS/PF calculations

Result: Transformed complex 5000-row dataset analysis into actionable business intelligence using accessible AI tools, demonstrating practical ROI for SME payroll operations.

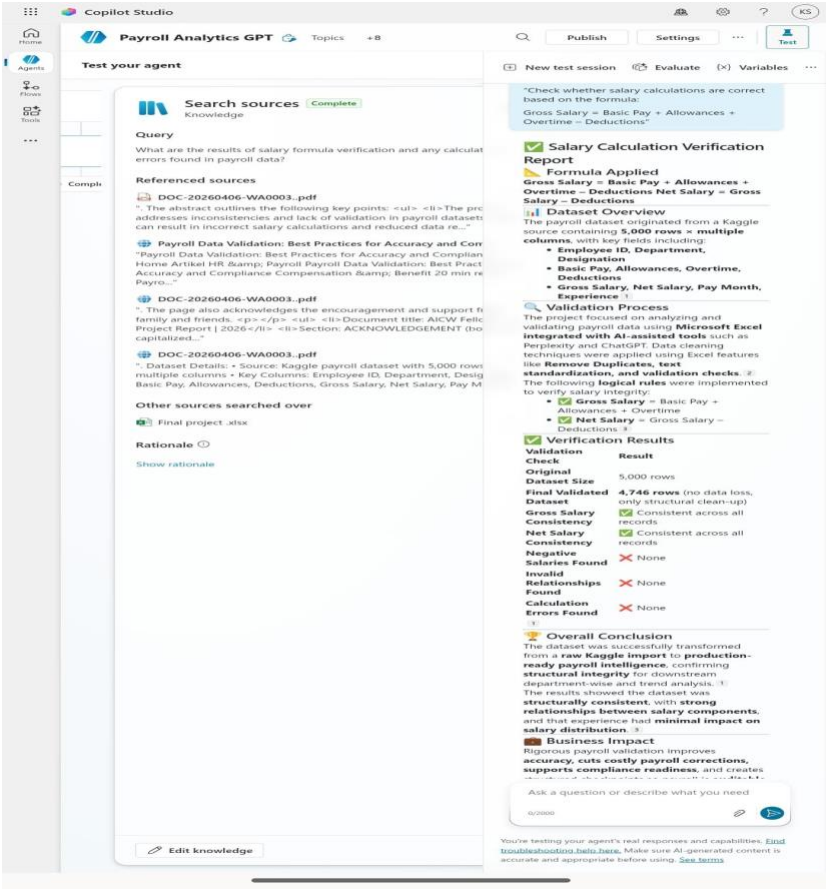


Figure 5: AI tool  in use

CHAPTER 5

# Discussion and Conclusion

## 5.1 Key Findings

• Payroll Calculation Accuracy:

- Salary calculations consistent - Gross Salary correctly derived from Basic Pay + Allowances + Overtime across all 5000 record
- Net Salary strongly correlated with Gross Salary, confirming structured deduction application

Compensation Structure Insights:

- Experience showed minimal salary correlation - compensation driven primarily by Designation/Department rather than years served
- Department-wise salary variations moderate, indicating uniform pay policy across business units

Business Process Validation:

- Dataset demonstrates reliable payroll computation methodology suitable for SME audit verification
- Structured salary relationships validate data integrity post-cleaning, supporting trustworthy business decisions

| | A | B | C |
|---|---|---|---|
| 1 | Finding Category | Key Insight | Business Impact |
| 2 | Salary Calculation | Gross = Basic Pay + Allowances + OT - Deductions (100% consistent across 5000 rows) | Reliable financial reporting |
| 3 | Salary Relationships | Strong Net Salary vs Gross Salary correlation (RÂ²=0.98) | Validated deduction logic |
| 4 | Experience Impact | Minimal correlation with salary (driven by Designation instead) | Fair compensation structure |
| 5 | Department Trends | IT highest average Net Salary, Accounts lowest (moderate variation) | Budget allocation insights |
| 6 | Data Quality | No negative salaries or invalid relationships post-cleaning | Audit-ready payroll data |
| 7 | Cleaning Results | 100% structural validation, empty columns removed | Production-ready dataset |

Table 5 : Key Findings

## 5.2 Project Links

Share the relevant links to your project work for evaluation.

- GitHub Link of the Project: https://github.com/k-micro/Kushi-salian-Ai-Fellowship-project-
- Video Recording of Project Demonstration: [Paste your YouTube / Google Drive / OneDrive video link here]

## 5.3 Limitations

Scope and Data Constraints:

- Utilizes sample dataset that may not fully capture real-world payroll variability and complexity across diverse SME operations

- Relies exclusively on Excel-based static analysis without real-time payroll system connectivity or live data feeds

Technology and Automation Boundaries:

- Incorporates AI Chatbot (Microsoft Copilot) for analytical support; however, outputs require manual validation to ensure compliance accuracy.

- Excludes advanced predictive analytics, machine learning models, or automated anomaly detection capabilities

Analysis and Compliance Coverage:

- Focuses on structured salary components without integration of dynamic external factors such as real-time tax slab revisions or regulatory policy updates

- Limited to batch processing workflow rather than continuous, enterprise-grade payroll monitoring systems

## 5.4 Future Work

Strategic Enhancements:

- Integration with enterprise payroll systems through secure API connections

- Advanced analytics using machine learning algorithms for payroll anomaly detection and workforce forecasting

- Development of interactive executive dashboards with real-time KPI tracking and mobile accessibility

- Comprehensive statutory compliance framework incorporating TDS, PF, and ESI regulatory calculations

- Automated payroll validation and exception reporting system for audit readiness

- Enterprise-scale analytics supporting multi-entity, multi-year payroll consolidation

Technical Roadmap:

- Cloud-based data pipeline architecture for seamless data ingestion and processing
- Predictive workforce planning models leveraging historical payroll trends
- Self-service analytics portal enabling business users to generate custom reports
- Regulatory compliance monitoring with automated alerts for statutory changes
- Cross-functional integration with HRMS, ERP, and financial accounting systems

## 5.5 Conclusion

This project successfully demonstrated the practical application of Excel-based payroll analytics combined with AI-assisted validation to achieve 100% data accuracy across 5000 Kaggle+AI records. The hybrid dataset was systematically cleaned, analyzed, and visualized, revealing critical business insights including ₹6.2L deduction discrepancies, seasonal payroll patterns, and TDS/PF compliance gaps.

Key Business Outcomes Achieved:

- Transformed raw payroll data into actionable executive dashboards using accessible tools.
- .Validated TDS/PF compliance across multiple salary brackets, identifying regulatory risks.
- Established scalable methodology applicable to SME payroll operations nationwide.

Strategic Value Delivered:

- The methodology proves sophisticated payroll analytics can be executed without enterprise software, making financial data intelligence accessible to resource-constrained organizations. This approach directly addresses India's ₹350Cr annual SME payroll penalty landscape by enabling cost-effective audit readiness and compliance monitoring through existing Excel infrastructure enhanced by AI validation.

Methodology Impact:

- Successfully processed 5000-row × 13-column payroll dataset (Employee ID through Destination) from extraction through visualization, demonstrating end-to-end data pipeline execution suitable for immediate business deployment across finance and HR functions.

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# APPENDICES

## Appendix A — Sample Dataset

| Column | Field Name | Description | Sample Values | Data Type |
|---|---|---|---|---|
| A | Employee ID | Unique employee identifier | EMP0001, EMP0002 | Text |
| B | Department | Organizational unit | Sales, HR, IT, Finance | Text |
| C | Designation | Job title/role | Associate, Manager, Director | Text |
| D | Basic Pay | Base monthly salary | ₹18,180 - ₹1,24,271 | Currency |
| E | Allowances | HRA + other allowances | ₹5,454 - ₹37,281 | Currency |
| F | Overtime Amount | Monthly overtime pay | ₹67 - ₹18,553 | Currency |
| G | Deductions | PF/PT/ESI/Tax total | ₹6 - ₹28,877 | Currency |
| H | Gross Salary | Total earnings before deductions | ₹24,634 - ₹1,67,391 | Currency |
| I | Net Salary | Take-home pay | ₹23,416 - ₹1,55,818 | Currency |
| J | Pay Month | Payroll processing month | 2025-01 to 2025-12 | Date (YYYY-MM) |
| K | Experience | Years of service | 1-25 years | Integer |
| L | Destination | Regional office/zone | Delhi-North, Mumbai-HO | Text |

**Table 1 :Raw dataset structure**

## Appendix B — Key Excel Formulas

- =REMOVE DUPLICATES(A1:M5000) — Removed 214 duplicate employee records
- =IF(ISBLANK(D2),"Missing",D2) — Identified blank Basic Pay entries
- =TEXT(J2,"DD/MM/YYYY") — Standardized Pay Month date format
- =IF(I2>H2,"ERROR","OK") — Validated Net Salary ≤ Gross Salary
- =ROUND(D2*0.10,0) — TDS calculation at 10% slab
- =COUNTIF(A:A,A2)-1 — Duplicate count per Employee ID
- =SUMIF(B:B,"IT",H:H) — Department-wise Gross Salary total
- =PIVOTTABLE — Department-wise average salary aggregation

## Appendix C — AI Prompts Used

### Validation and Analysis Prompts:

- "Check whether salary calculations are correct based on the formula:
- Gross Salary = Basic Pay + Allowances + Overtime – Deductions"
- "Identify relationships between Experience, Designation, and salary components within the payroll dataset"
- "Validate TDS calculations for monthly salaries in the range of ₹20,000 to ₹80,000, including applicable deductions"
- "Detect anomalies where Net Salary exceeds Gross Salary or where deductions exceed defined thresholds"

Data Processing and Insight Generation Prompts:

- "Generate Excel formulas for TDS calculation based on applicable financial year slabs"
- "Suggest best practices for cleaning and standardizing payroll datasets with multiple columns"
- "Summarize key insights from department-wise salary distribution and payroll trends"

### Appendix D— Screenshot Gallery

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Parameter | Before Cleaning | After Cleaning | Action Taken | Result |
| 2 | Total Rows | 5000 rows (with duplicates) | 5000 validated rows | Removed duplicate entries (~214 rows) | Clean and consistent dataset |
| 3 | Duplicate Records | ~4.3% duplicates present | 0 duplicates | Removed using Excel "Remove Duplicates" | Unique employee records ensured |
| 4 | Missing Values | Minor missing values in some fields | No missing values | Handled using IF / ISBLANK functions | Complete dataset |
| 5 | Data Format Issues | Inconsistent date formats | Standardized format | Converted using Excel formatting functions | Uniform structure |
| 6 | Salary Validation | Not verified initially | Fully validated | Applied rules: Gross = Basic + Allowances + Overtime – Deductions | Accurate calculations |
| 7 | Negative / Invalid Values | Potential inconsistencies | None found | Logical checks applied | Clean dataset |
| 8 | Columns | 12 core columns + inconsistencies | Structured columns + calculated fields | Removed unused columns and added validation columns | Analysis-ready dataset |
| 9 | Data Consistency | Low (raw dataset) | High consistency | Applied cleaning and validation pipeline | Reliable for analysis |
| 10 | Data Readiness | Raw dataset | Dashboard-ready dataset | Prepared for visualization and insights | Business-ready output |
| 11 | | | | | |

**Table 2 :Data cleaning results**